

Linear Code from a Generator Matrix

Encoding $E : (\mathbb{Z}/2)^3 \rightarrow (\mathbb{Z}/2)^5$

Generator Matrix and Encoding

We are given the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

We denote its rows by

$$r_1 = (1, 0, 0, 0, 1), \quad r_2 = (0, 0, 0, 1, 1), \quad r_3 = (0, 1, 1, 1, 0).$$

The encoding function E is defined by

$$E(w) = wG \quad \text{over } \mathbb{Z}_2,$$

where $w = (w_1, w_2, w_3) \in (\mathbb{Z}/2)^3$ is a row vector, and the resulting codeword $c = (c_1, \dots, c_5) \in (\mathbb{Z}/2)^5$ is obtained by matrix multiplication modulo 2:

$$c_j = w_1 G_{1j} + w_2 G_{2j} + w_3 G_{3j} \pmod{2}, \quad j = 1, \dots, 5.$$

Equivalently, since we are in $\mathbb{Z}_2$,

$$E(w) = w_1 r_1 \oplus w_2 r_2 \oplus w_3 r_3$$

where $\oplus$ is bitwise addition modulo 2.

There are $2^3 = 8$ possible messages w , so there are 8 codewords in total.

(a) Compute All Codewords and the Minimum Distance

We list all messages $w \in (\mathbb{Z}/2)^3$ and compute their corresponding codewords $c = E(w)$ explicitly.

Message $w = (0, 0, 0)$

$$E(000) = 0 \cdot r_1 \oplus 0 \cdot r_2 \oplus 0 \cdot r_3 = (0, 0, 0, 0, 0).$$

Message $w = (0, 0, 1)$

$$E(001) = 0 \cdot r_1 \oplus 0 \cdot r_2 \oplus 1 \cdot r_3 = r_3 = (0, 1, 1, 1, 0).$$

So

$$E(001) = 01110.$$

Message $w = (0, 1, 0)$

$$E(010) = 0 \cdot r_1 \oplus 1 \cdot r_2 \oplus 0 \cdot r_3 = r_2 = (0, 0, 0, 1, 1).$$

So

$$E(010) = 00011.$$

Message $w = (0, 1, 1)$

$$E(011) = 0 \cdot r_1 \oplus 1 \cdot r_2 \oplus 1 \cdot r_3 = r_2 \oplus r_3.$$

Compute $r_2 \oplus r_3$ bit by bit:

	1	2	3	4	5
r_2	0	0	0	1	1
r_3	0	1	1	1	0
$r_2 \oplus r_3$	0	1	1	0	1

Thus

$$E(011) = 01101.$$

Message $w = (1, 0, 0)$

$$E(100) = 1 \cdot r_1 \oplus 0 \cdot r_2 \oplus 0 \cdot r_3 = r_1 = (1, 0, 0, 0, 1).$$

So

$$E(100) = 10001.$$

Message $w = (1, 0, 1)$

$$E(101) = 1 \cdot r_1 \oplus 0 \cdot r_2 \oplus 1 \cdot r_3 = r_1 \oplus r_3.$$

Compute $r_1 \oplus r_3$:

	1	2	3	4	5
r_1	1	0	0	0	1
r_3	0	1	1	1	0
$r_1 \oplus r_3$	1	1	1	1	1

Thus

$$E(101) = 11111.$$

Message $w = (1, 1, 0)$

$$E(110) = 1 \cdot r_1 \oplus 1 \cdot r_2 \oplus 0 \cdot r_3 = r_1 \oplus r_2.$$

Compute $r_1 \oplus r_2$:

	1	2	3	4	5
r_1	1	0	0	0	1
r_2	0	0	0	1	1
$r_1 \oplus r_2$	1	0	0	1	0

Thus

$$E(110) = 10010.$$

Message $w = (1, 1, 1)$

$$E(111) = 1 \cdot r_1 \oplus 1 \cdot r_2 \oplus 1 \cdot r_3 = r_1 \oplus r_2 \oplus r_3.$$

We already computed $r_1 \oplus r_2 = (1, 0, 0, 1, 0)$. Now add r_3 :

	1	2	3	4	5
$r_1 \oplus r_2$	1	0	0	1	0
r_3	0	1	1	1	0
$(r_1 \oplus r_2) \oplus r_3$	1	1	1	0	0

Thus

$$E(111) = 11100.$$

Summary of all codewords

w	$E(w)$
000	00000
001	01110
010	00011
011	01101
100	10001
101	11111
110	10010
111	11100

So the code $\mathcal{C}$ is:

$$\mathcal{C} = \{00000, 01110, 00011, 01101, 10001, 11111, 10010, 11100\}.$$

Minimum distance via Hamming weight

For a *linear* code, the distance between two codewords c and c' is

$$d(c, c') = \text{wt}(c - c').$$

Over $\mathbb{Z}_2$, subtraction is the same as addition, so

$$d(c, c') = \text{wt}(c \oplus c').$$

But if c and c' are codewords, then $c \oplus c'$ is also a codeword (closure under addition). So:

$$l_{\min} = \min_{c \neq c'} d(c, c') = \min_{c \neq 0} \text{wt}(c).$$

Therefore we just need the smallest Hamming weight among the non-zero codewords. Compute weights:

Codeword	Weight (number of 1's)
01110	3
00011	2
01101	3
10001	2
11111	5
10010	2
11100	3

The smallest non-zero weight is 2 (achieved by 00011, 10001, and 10010), so

$$\boxed{l_{\min} = 2}.$$

(b) Find a Non-zero Codeword

In this part, we are asked to find a *non-zero* codeword for the encoding E .

What is a codeword?

A **codeword** is any vector of the form

$$c = E(w) = wG,$$

where $w \in (\mathbb{Z}/2)^3$ is a 3-bit message and G is the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

We think of w as a row vector:

$$w = (w_1, w_2, w_3),$$

and the encoding rule is

$$E(w) = w_1 r_1 \oplus w_2 r_2 \oplus w_3 r_3,$$

where r_1, r_2, r_3 are the rows of G , and $\oplus$ denotes bitwise addition modulo 2 (XOR).

What does “non-zero codeword” mean?

The bitstring consisting only of zeros is

$$00000 = (0, 0, 0, 0, 0).$$

The problem explicitly says: “any codeword will do, as long as it is not the bitstring consisting only of zeros.” So we must find some message w such that:

- $E(w)$ is a codeword, and
- $E(w) \neq 00000$.

Because this is a **linear** code:

- The message $w = (0, 0, 0)$ always gives $E(000) = 00000$. - Any *other* message $w \neq (0, 0, 0)$ will give a non-zero codeword (since the rows of G are linearly independent).

So *any non-zero message* is a valid starting point.

Choosing a simple non-zero message

There are 7 non-zero messages in $(\mathbb{Z}/2)^3$:

$$001, 010, 011, 100, 101, 110, 111.$$

We can choose any of these. For simplicity, we choose

$$w = (1, 0, 0),$$

because this only “picks out” the first row of G , making the computation very simple.

Computing the codeword for $w = (1, 0, 0)$

Using the encoding rule

$$E(w) = w_1 r_1 \oplus w_2 r_2 \oplus w_3 r_3,$$

we plug in $w = (1, 0, 0)$:

$$E(100) = 1 \cdot r_1 \oplus 0 \cdot r_2 \oplus 0 \cdot r_3.$$

Now compute step by step:

$$1 \cdot r_1 = r_1 = (1, 0, 0, 0, 1),$$

$$0 \cdot r_2 = (0, 0, 0, 0, 0),$$

$$0 \cdot r_3 = (0, 0, 0, 0, 0).$$

Adding them modulo 2 (bitwise XOR):

$$E(100) = r_1 \oplus (0, 0, 0, 0, 0) \oplus (0, 0, 0, 0, 0) = r_1 = (1, 0, 0, 0, 1).$$

Written as a bitstring, this is:

$$E(100) = 10001.$$

Check that it is non-zero

The codeword 10001 is clearly not the all-zero word 00000, so it satisfies the condition of being a *non-zero* codeword.

Therefore, a correct answer for part (b) is:

$$\boxed{10001}.$$

Remark: Other valid answers

Any other non-zero message would also give a valid non-zero codeword. For example:

$$E(001) = 01110,$$

$$E(010) = 00011,$$

$$E(011) = 01101,$$

$$E(101) = 11111,$$

$$E(110) = 10010,$$

$$E(111) = 11100.$$

Any of these bitstrings could also be used as the answer to part (b). We chose 10001 only because it is the simplest to compute.

(c) A 5-bit String that is Not a Codeword

All codewords of this code are:

00000, 01110, 00011, 01101, 10001, 11111, 10010, 11100.

We just need any 5-bit string that does *not* appear in this list. For example,

01010

is not among the codewords, so it is not a codeword of this code.

Thus a valid answer is:

01010.

Final Answers (in bitstring form)

- (a) $l_{\min} = 2$
- (b) Example of a non-zero codeword: 10001
- (c) Example of a non-codeword: 01010